

Prashanth Durgam

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PROFESSIONAL SUMMARY

M.Tech CSE graduate from NIT Durgapur with experience in full-stack development and AI-driven systems. Strong foundation in software engineering, algorithms, and problem solving, with research experience in AI, IoT, and intelligent automation.

EDUCATION

•National Institute of Technology, Durgapur	2024 - 2026
Master of Technology (Computer Science and Engineering)	CGPA: 9.00
•Chaitanya Bharathi Institute Of Technology, Hyderabad	2019 - 2023
Bachelor of Engineering (Electronics and Communication Engineering)	CGPA: 8.09
•Sri Chaitanya Junior Kalasala, Hyderabad	2017 - 2019
Telangana State Board of Intermediate Education, Telangana	Percentage: 95.9

PROJECTS

- Automated Personal Comfort Recommendation System | Generative AI Model
M.Tech Major Project
 - **Tools & Technologies:** ReactJS, ExpressJS, Python, FastAPI, Gemini 2.5-flash, and Pydantic.
 - Developed a full-stack AI system integrating a comfort recommendation engine with an interactive Gemini-powered conversational assistant.
 - Engineered dynamic appliance control heuristics using real-time indoor, outdoor, and environmental sensor telemetry.
 - Implemented a user feedback loop and schema validation using Pydantic to enhance recommendation precision and response reliability.
- Document Q&A Chatbot | Retrieval-Augmented Generation (RAG)
Generative AI Application
 - **Tools & Technologies:** Python, LangChain, Google Gemini, Streamlit, PyPDF, Vector Embeddings.
 - Built an end-to-end RAG chatbot enabling real-time, context-grounded Q&A on user-uploaded PDF documents.
 - Implemented document chunking, semantic vector indexing, and similarity search using Google Gemini Embeddings and LangChain.
 - Developed a responsive Streamlit chat interface with multi-turn conversation memory and dynamic prompt orchestration.

PUBLICATIONS

- AirWise: LLM-Powered Smart Air Quality Monitoring and Automated Appliance Control, *AIEE 2026*.
- iAirGuard: A Modular IoT Architecture with Dynamic Sampling and TinyML-Based Fault Detection for Air Quality Monitoring, *IEEE COMSNETS 2026*.

TECHNICAL SKILLS AND INTERESTS

Languages: Java, Python, C, JavaScript
Web Tech: HTML, CSS, React.js, Node.js, Express.js, FastAPI, REST APIs
AI & DB: LLM, LangChain, RAG, Agentic AI, SQL, MongoDB
Tools & Core: Git, GitHub, VS Code, Data Structures & Algorithms, OOP
Spoken: English, Hindi, Marathi, Telugu

CERTIFICATES & ACHIEVEMENTS

- Qualified GATE 2024
- Python Programming: Complete Python Bootcamp (Udemy)
- SQL Fundamentals: SQL Certification (CodeChef)